

Test Statistic Calculation

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For tests of a *Population Mean/Average*, the z-statistic is

$$z = \frac{\left(\begin{smallmatrix} \text{sample} \\ \text{mean} \end{smallmatrix}\right) - \left(\begin{smallmatrix} \text{population mean} \\ \text{under } H_0 \end{smallmatrix}\right)}{SE}, \quad \text{where } SE = \frac{\text{sample SD}}{\sqrt{\text{sample size}}}.$$

For tests of a *population proportion*, the z-statistic is

$$z = \frac{\left(\begin{smallmatrix} \text{sample} \\ \text{percentage} \end{smallmatrix}\right) - \left(\begin{smallmatrix} \text{population percentage} \\ \text{under } H_0 \end{smallmatrix}\right)}{SE}$$

where

$$SE = \frac{\sqrt{p_0(1-p_0)}}{\sqrt{\text{sample size}}} \times 100\%, \quad \text{where } p_0 = \text{proportion of } \boxed{1} \text{ in box under } H_0 \\ = \text{population proportion under } H_0.$$

Exercise 1: Test Statistic of Population *Mean*

To test H_0 : Population mean = 25 v.s. H_a : Population mean $\neq$ 25, we take a simple random sample from the population and get

- ▶ sample size = 201
- ▶ sample mean = 21.9
- ▶ sample SD = 18.0

Find the z-statistic.

Ans:

Exercise 1: Test Statistic of Population *Mean*

To test H_0 : Population mean = 25 v.s. H_a : Population mean $\neq$ 25, we take a simple random sample from the population and get

- ▶ sample size = 201
- ▶ sample mean = 21.9
- ▶ sample SD = 18.0

Find the z-statistic.

Ans:

$$SE = \frac{\text{sample SD}}{\sqrt{\text{sample size}}} = \frac{18.0}{\sqrt{201}} \approx 1.27,$$
$$z = \frac{\left(\begin{smallmatrix} \text{sample} \\ \text{mean} \end{smallmatrix}\right) - \left(\begin{smallmatrix} \text{population mean} \\ \text{under } H_0 \end{smallmatrix}\right)}{SE} = \frac{21.9 - 25}{1.27} \approx -2.44.$$

Exercise 2: Test Statistic of Population *Mean*

To test H_0 : Population mean = 3.0 v.s. H_a : Population mean $>$ 3.0, we take a simple random sample from the population and get

- ▶ sample size = 168
- ▶ sample mean = 4.2
- ▶ sample SD = 5.6

Find the z-statistic.

Ans:

Exercise 2: Test Statistic of Population *Mean*

To test H_0 : Population mean = 3.0 v.s. H_a : Population mean > 3.0 , we take a simple random sample from the population and get

- ▶ sample size = 168
- ▶ sample mean = 4.2
- ▶ sample SD = 5.6

Find the z-statistic.

Ans:

$$SE = \frac{\text{sample SD}}{\sqrt{\text{sample size}}} = \frac{5.6}{\sqrt{168}} \approx 0.432,$$
$$z = \frac{\left(\begin{smallmatrix} \text{sample} \\ \text{mean} \end{smallmatrix}\right) - \left(\begin{smallmatrix} \text{population mean} \\ \text{under } H_0 \end{smallmatrix}\right)}{SE} = \frac{4.2 - 3.0}{0.432} \approx 2.78.$$

Exercise 3: Test Statistic of Population *Percentage*

To test H_0 : Population percentage = 12% v.s. H_a : Population percentage < 12%, we take a simple random sample from the population and get

- ▶ sample size = 570
- ▶ sample percentage = 9.2%

Find the z-statistic.

Ans:

Exercise 3: Test Statistic of Population *Percentage*

To test H_0 : Population percentage = 12% v.s. H_a : Population percentage < 12%, we take a simple random sample from the population and get

- ▶ sample size = 570
- ▶ sample percentage = 9.2%

Find the z-statistic.

Ans:

$$SE = \frac{\sqrt{p(1-p)}}{\sqrt{\text{sample size}}} \times 100\% = \frac{\sqrt{0.12(1-0.12)}}{\sqrt{570}} \times 100\% = 0.0136 \times 100\% = 1.36\%$$
$$z = \frac{\left(\begin{smallmatrix} \text{sample} \\ \text{percentage} \end{smallmatrix}\right) - \left(\begin{smallmatrix} \text{population percentage} \\ \text{under } H_0 \end{smallmatrix}\right)}{SE} = \frac{9.2\% - 12\%}{1.36\%} \approx -2.06.$$

Exercise 4: Test Statistic of Population *Percentage*

To test H_0 : Population percentage = 80% v.s. H_a : Population percentage $\neq$ 80%, we take a simple random sample from the population and get

- ▶ sample size = 1400
- ▶ sample percentage = 83.5%

Find the z-statistic.

Ans:

Exercise 4: Test Statistic of Population *Percentage*

To test H_0 : Population percentage = 80% v.s. H_a : Population percentage $\neq$ 80%, we take a simple random sample from the population and get

- ▶ sample size = 1400
- ▶ sample percentage = 83.5%

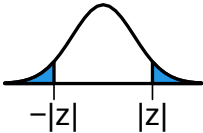
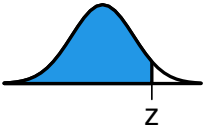
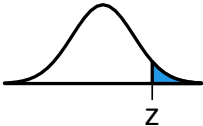
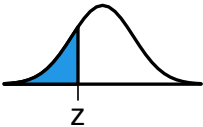
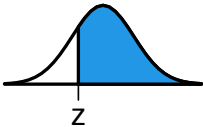
Find the z-statistic.

Ans:

$$SE = \frac{\sqrt{p(1-p)}}{\sqrt{\text{sample size}}} \times 100\% = \frac{\sqrt{0.8(1-0.8)}}{\sqrt{1400}} \times 100\% = 0.0107 \times 100\% = 1.07\%$$
$$z = \frac{\left(\begin{array}{c} \text{sample} \\ \text{percentage} \end{array} \right) - \left(\begin{array}{c} \text{population percentage} \\ \text{under } H_0 \end{array} \right)}{SE} = \frac{83.5\% - 80\%}{1.07\%} \approx 3.27.$$

P -Value Calculation

Summary of the One-Tailed and Two-Tailed Tests

	two-sided	lower one-sided	upper one-sided
H_0 H_A	Ave. of box = μ Ave. of box $\neq \mu$	Ave. of box = μ Ave. of box $< \mu$	Ave. of box = μ Ave. of box $> \mu$
P-value when $z > 0$			
P-value when $z < 0$			

Exercise 1: P-Value

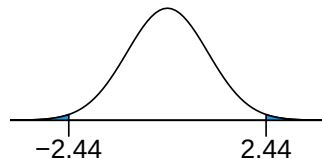
To test H_0 : Population mean = 25 v.s. H_a : Population mean $\neq$ 25, recall we get the z -statistic

$$z \approx -2.44.$$

Find the P -value.

Ans:

H_a is two-sided ($\neq$), the lower-tail area below -2.44 is 0.0073.



$$\text{Two-sided } P\text{-value} = 2 \times 0.0073 = 0.0146.$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064

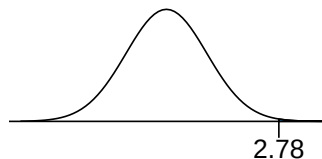
Exercise 2: P-value

To test H_0 : Population mean = 3.0 v.s. H_a : Population mean $>$ 3.0, recall we get the z -statistic

$$z \approx 2.78.$$

Find the P -value.

H_a is upper one-sided ($>$), the lower-tail area below 2.78 is 0.9973.



$$\text{Upper one-sided } P\text{-value} = 1 - 0.9973 = 0.0027.$$

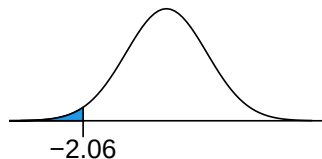
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974

Exercise 3: P-value

To test H_0 : Population percentage = 12% v.s. H_a : Population percentage < 12%, recall we get the z -statistic

$$z \approx -2.06$$

Find the P -value.



H_a is lower one-sided ($<$), the lower-tail area below -2.06 is 0.0197.

lower one-sided P -value = 0.0197.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183

Exercise 4: P -value

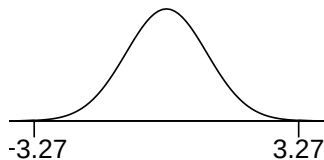
To test H_0 : Population percentage = 80% v.s. H_a : Population percentage $\neq$ 80%, recall we get the z -statistic

$$z \approx 3.27.$$

Find the P -value.

H_a is two-sided ($\neq$), the lower-tail area below 3.27 is 0.9995.

$$\text{two-sided } P\text{-value} \approx 2(1 - 0.9995) = 0.001$$



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995

Example (Blood Donors)

We wish to make inference on the percentage of regular blood donors in a U.S. town, in which 10,000 of the residents are age-eligible to donate blood. To that end, we take a random sample of 400 age-eligible people from the town and find that 22 of them have donated blood within the past 12 months.

To analyze the survey, first set up a box model.

- ▶ There is one ticket in the box for
- ▶ How many tickets in the box?
- ▶ On the ticket shows
- ▶ How many draws?

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To analyze the survey, first set up a box model.

- ▶ There is one ticket in the box for each people age-eligible to donate blood in the town
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To analyze the survey, first set up a box model.

- ▶ There is one ticket in the box for each people age-eligible to donate blood in the town
- ▶ How many tickets in the box? 10000
- ▶ On the ticket shows
- ▶ How many draws?

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To analyze the survey, first set up a box model.

- ▶ There is one ticket in the box for each people age-eligible to donate blood in the town
- ▶ How many tickets in the box? 10000
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- ▶ How many draws?

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To analyze the survey, first set up a box model.

- ▶ There is one ticket in the box for each people age-eligible to donate blood in the town
- ▶ How many tickets in the box? 10000
- ▶ On the ticket shows 1 if he/she has donated blood in the past 12 months, and 0 otherwise
- ▶ How many draws? 400 = sample size

Example (Blood Donors, Continued)

According to American Red Cross, only 3% of age-eligible people in the U.S. donate blood annually. We wish to test whether the town's percentage of blood donors is different from the nationwide percentage.

- a. Clearly state the null and alternative hypotheses.
 - ▶ H_0 : The percentage of blood donors in the town is 3%
 - ▶ H_A : The percentage of blood donors in the town is not 3%
- b. Calculate the appropriate test statistic.

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- a. Clearly state the null and alternative hypotheses.
 - ▶ H_0 : The percentage of blood donors in the town is 3%
 - ▶ H_A : The percentage of blood donors in the town is not 3%
- b. Calculate the appropriate test statistic.
 - ▶ sample percentage = $\frac{22}{400} \times 100\% = 5.5\%$;
 - ▶ population percentage under H_0 is 3%.
 - ▶ sample size = 400

$$SE = \frac{\sqrt{p(1-p)}}{\sqrt{\text{sample size}}} \times 100\% = \frac{\sqrt{0.03(1-0.03)}}{\sqrt{400}} \times 100\% = 0.00853 \times 100\% = 0.853\%$$
$$z = \frac{\left(\begin{smallmatrix} \text{sample} \\ \text{percentage} \end{smallmatrix}\right) - \left(\begin{smallmatrix} \text{population percentage} \\ \text{under } H_0 \end{smallmatrix}\right)}{SE} = \frac{5.5\% - 3\%}{0.853\%} \approx 2.93$$

Example (Blood Donors, Continued)

c. Find the P -value.

- ▶ From the normal table, the lower-tail area corresponds to $z = 2.93$ is about 0.9983. The two-sided P -value is thus $2(1 - 0.9983) = 0.0034$.

d. Is the percentage of blood donors in the town significantly different from the nationwide percentage, at 1% level?

- ▶ Yes, since the P -value 0.0034 is below the $1\% = 0.01$ significance level.